MIL-C-915/42A
1 August 1972
SUPERSEDING 1/
MIL-C-915/42(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 1S50MU

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, ASTM B-286 size 22-7, tin coated. Maximum diameter 0.033 inch.

(A) Second - Polyethylene insulation, white, type II, class L, grade 4 of L-P-390. Nominal thickness 0.027 inch.

(A) Third - Clear polyamide jacket on each conductor type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch. Standard Identification Code, applied by method 1.

Fourth - Braided shield of AWG No. 36 tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.

(A) Fifth - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness of 0.003 inch. Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed.

(A) Sixth - The required number of singles (see details) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from center outward.

Seventh - Binder tape applied helically with overlap.

(A) Eighth - Standard thermoplastic jacket (gray). (Nominal thickness 0.050 inch.) Cable surface marking required.

Details

(A)

Type and size	Number of condrs.	Diameter over jacket condr. nom. (inch)	Overall diameter		Cold (A) bending mandrel diameter maximum (inches)
			Min. (inches)	Max. (inches)	
1S50MU-16	16	0.131	0.760	0.825	11
1S50MU-20	20	.131	.835	.905	12
1S50MU-40	40	.131	1.095	1.185	15
1S50MU-70	70	.131	1.465	1.555	20

EXAMINATION AND TESTS:

	Requirements:
(A) Basic electrical:	
Conductor resistance - At 25°C., ohms/1000 ft., maximum.	17.54
Voltage withstand - Volts, rms, minimum	
Conductor to shield	5000
Shield to shield.	500

1/ Requirements formerly covered on MIL-C-24145/2.

(A) Denotes changes

FSC 6145

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	Requirements:
Group A:	
Visual and dimensional	
Characteristic impedance - At 1 MHz, ohms.	50 ± 2
Group B:	
Attenuation - At 400 MHz, db/100 ft., maximum.	18
(A) Cold bending, cable - At 0° ± 2°C., (See Details for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum.	1400
Elongation - Percent, minimum.	300
Jacket (cable)	
Tensile strength - psi, minimum.	1800
Elongation - Percent, minimum.	225
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - Percent of unaged, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (conductor) - Cycles, minimum.	50
Permanence of printing (jacket) - Cycles, minimum.	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS:

All sizes: 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609-30)

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